EventTrail - Smart Campus Event Discovery & Navigation Platform

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INTRODUCTION

- University campuses host hundreds of events, but event details are scattered across WhatsApp, noticeboards, and emails.
- The absence of a formal RSVP system makes it difficult to predict attendance or manage venue capacity
- EVENTTRAIL - is a full-stack, serverless web platform designed to centralise event discovery and management.
- The platform integrates an interactive campus map that guides students step-by-step from their current location to the exact event room.
- It provides dedicated tools for students to discover activities and for clubs to manage their memberships.
- The ultimate goal is to boost event attendance, reduce administrative workloads, and strengthen campus community bonds

LITERATURE REVIEW

Paper 1:

Title: Eventify: Campus Event Management Web-App

Author(s): Sandeep Mehta, K. Srikar Goud, et al. (ISJEM, 2024)

Methodology: Developed a web-based platform with role-based access for administrators, organizers, and students, featuring an online event approval workflow.

Advantages: Centralizes the event tracking process, improves overall event visibility, and significantly reduces the manual workload of administrative staff

Limitations: Earlier system iterations lacked robust mobile integration and did not support real-time push notifications for immediate student updates.

Link: https://ijirt.org/publishedpaper/IJIRT186834_PAPER.pdf

LITERATURE REVIEW

Paper 2:

- **Title:** Smart Campus Event Hub
- **Author(s):** P.Saraswathy, Ch.Madurai, D.Mythil,IJRASET Research Publication (2024)
- **Methodology:** An intelligent platform developed with Flask and AI components that generates event recommendations by analyzing user profiles alongside academic timetables.
- **Advantages:** Actively prevents scheduling conflicts, utilizes students' free time effectively, and increases engagement through automated, personalized alerts.
- **Limitations:** Requires complex integration with dynamic student academic timetables and relies heavily on constant data updates to remain accurate.
- **Link:** https://www.ijraset.com/best-journal/smart-campus-event-hub

LITERATURE REVIEW

- **Paper 3:**
- **Title:** Adaptive QR Campus Navigation System
- **Author(s):** Dr. Archana Potnurwar, Saurav Chafle, Yash Malode, Tejas Gahane, Mayur Nerpargar, Chetana Ambulkar (IJIRT, 2025)
- **Methodology:** QR-based web navigation using Firebase, Leaflet.js/Mapbox, and Dijkstra/A* algorithms for indoor and outdoor routes.
- **Advantages:** Provides accurate indoor/outdoor navigation, step-by-step guidance, and supports voice and multilingual features.
- **Limitations:** Requires physical QR-code scanning, depends on internet connectivity, and requires proper placement and maintenance of QR codes across the campus.
- **Link:** https://ijirt.org/publishedpaper/IJIRT188859_PAPER.pdf

Research Gap Identified:

- **No Unified Platform:** Existing systems solve only one piece of the problem — event management, recommendations, or navigation — never all three together.
- **High Hosting Costs:** Existing systems lack an auto-scaling, serverless architecture to minimize idle costs.
- **No Auto-Waitlist:** Current platforms fail to automatically promote waitlisted users when seats open up.
- **Missing Tools & Alerts:** A lack of real-time analytics for admins and automated SMS/email reminders for students.
- **Lack of Integrated Navigation:** There is no unified system that handles real-time event updates while simultaneously providing map-based outdoor walking routes and indoor floor-plan guidance.

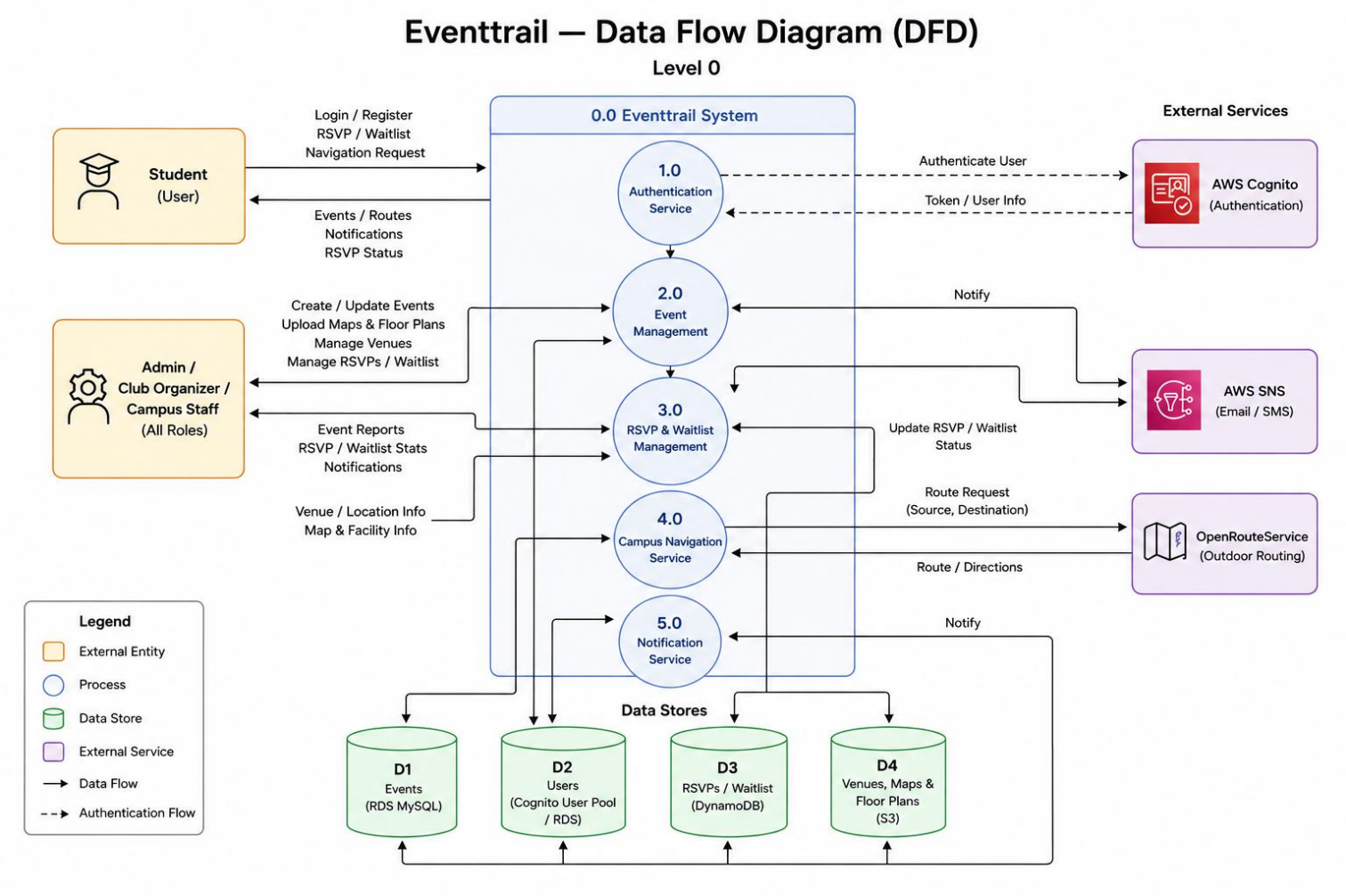
EXISTING SYSTEM

- **Fragmented Information:** No central platform; event details are scattered across channels.
- **Manual Processes:** Administrators rely on slow, error-prone spreadsheets for registration.
- **Poor Planning:** Organizers lack digital tools to predict attendance or manage venue capacity.
- **Navigation Barriers:** Students lack digital tools to locate venues, especially across multi-building, multi-floor campus layouts.
- **Negative Impact:** This manual approach leads to low attendance and increased student isolation.

PROPOSED SYSTEM

- **Centralized Architecture**: Built on an auto-scaling, serverless AWS infrastructure.
- **Centralized Hub:** Unified event feed with one-click RSVP and automated waitlist management.
- **Interactive Map:** Leaflet.js-based map displaying all event venues with real-time seat availability
- **Two-Phase Navigation**: Outdoor walking routes via OpenRouteService and indoor floor-plan overlays for room-level guidance.
- **Admin Dashboard:** Visual map editor for managing locations, floor plans, and venue changes without hardcoding.
- **Automated Alerts:** Built-in email and SMS reminders sent to students before their registered events.
- **Addresses the Gap:** Combines event management, real-time capacity control, and indoor+outdoor navigation in one serverless platform — the exact gap left by existing systems.

DATA FLOW DIAGRAM



METHODOLOGY

- **Architecture**: Fully serverless AWS design ensuring auto-scaling and zero maintenance.
- **Frontend:** React.js application hosted globally on Amazon S3 and delivered via CloudFront CDN.
- **Navigation:** Leaflet.js for mapping, OpenRouteService for outdoor pedestrian routing, and custom image overlays for floor plans.
- **Backend**: Node.js functions on AWS Lambda routed via API Gateway.
- **Databases**: MySQL (RDS) for relational data and DynamoDB for real-time RSVPs.
- **Security & Alerts**: AWS Cognito for user authentication and AWS SNS for notifications.

TECHNOLOGIES & TOOLS

- **Programming Languages:** JavaScript (ES6+), Node.js
- **Frameworks / Libraries:** React.js, Leaflet.js, Tailwind CSS
- **Database:** Amazon RDS (MySQL) for relational data, Amazon DynamoDB for real-time RSVP/waitlist state
- **Cloud & Platforms:** AWS Lambda, API Gateway, Amazon S3, CloudFront, AWS Cognito, Amazon SNS, Amazon EventBridge
- **External APIs:** OpenRouteService (outdoor pedestrian routing)
- **Dev & Design Tools:** Git/GitHub, Postman, Eraser.io (diagrams), Figma, Framer

SPRINT PLANNER

Sprint	Goal	PB No.	Deliverables
Sprint 1 – Authentication & Login	Implement secure user authentication and login functionality. SPRINT PLANNER	PB-01	AWS Cognito integration, Login page UI, Role-based authentication, User dashboard redirection.
Sprint 2 – Event Discovery & Event Management	Develop the event discovery module and enable administrators to manage events.	PB-02, PB-03	Event listing page, Search & filters, Event details page, Admin dashboard, Event CRUD operations.
Sprint 3 – RSVP & Waitlist Management	Implement event registration and automatic waitlist management.	PB-04	RSVP module, Capacity validation, Automatic waitlist management, Cancellation handling.
Sprint 4 – Campus Navigation	Implement navigation to event venues with indoor guidance.	PB-05	Interactive campus map, Outdoor routing, Geofence detection, Indoor step-by-step navigation.
Sprint 5 – Notifications & Attendance	Provide automated event notifications and attendance management.	PB-06	Amazon SNS notifications, Event reminders, Waitlist alerts, Venue update notifications, Attendance management.
Sprint 6 – System Integration & Deployment	Integrate all modules, test the system, and deploy the application.	—	Integrated EventTrail application, System testing, Bug fixes, AWS deployment, Final documentation.

EXPECTED OUTCOME AND CONCLUSION

- **Expected Results:** A fully functional, mobile-responsive web platform enabling real-time event discovery, RSVP, waitlist management, and door-to-room campus navigation.
- **Improved Attendance:** Automated reminders and waitlist promotion are expected to increase actual event turnout and eliminate wasted seats.
- **Reduced Admin Workload:** Club organizers gain a live dashboard, removing the need for manual spreadsheet-based tracking.
- **Real-World Application:** Directly deployable across university and college campuses to replace fragmented WhatsApp/notice-board based event communication.
- **Scalability:** The serverless AWS architecture scales automatically during peak RSVP periods and costs near-zero during quiet periods such as holidays.
- **Future Scope:** Develop dedicated Android and iOS applications for easier access to events, RSVPs, notifications, and campus information.

SPRINT 1 RESULT

- **Sprint Goal:** To implement a secure authentication system for EventTrail using AWS Cognito and provide a functional login interface for users.
- **Completed:**
  - AWS Cognito User Pool configured
  - Student / Admin authentication implemented
  - Login Page UI developed
  - Credential validation implemented
  - Role-based access enforced
  - Login redirection based on role
  - Authentication flow tested

SPRINT 2

- **Sprint Goal:** Develop the Event Discovery and Event Management features for both Student and Admin users. AWS Cognito User Pool configured
- **Current Status :** Sprint 2 is in progress. The UI for both Student and Admin has been created. The Admin event management backend, including CRUD operations, has been completed. The Student-side backend integration and event discovery functionality are still in progress.

THANK YOU